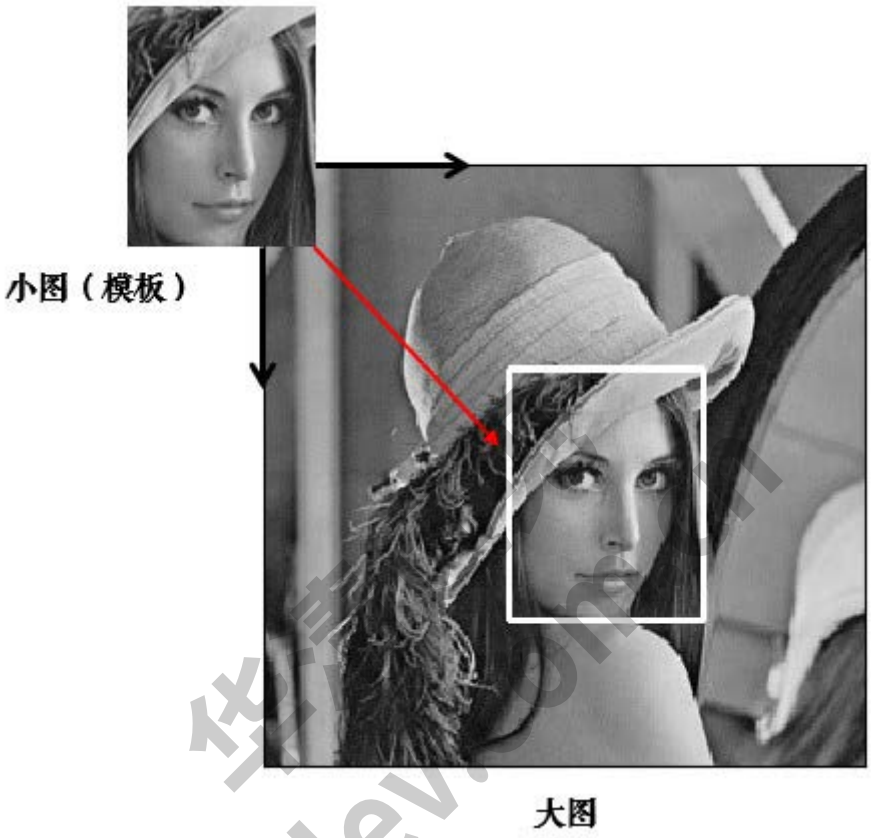
模板匹配

1. 模板匹配

模板匹配就是用模板图（通常是一个小图）在目标图像（通常是一个比模板图大的图片）中不断的滑动比较，通过某种比较方法来判断是否匹配成功。



2. 匹配方法

模板滑动比较的过程中有6种不同的比较方法。

2.1 平方差匹配TM_SQDIFF

以模板图与目标图所对应的像素值使用平方差公式来计算，其结果越小，代表匹配程度越高，计算过程举例如下。

注意：模板匹配过程皆不需要边缘填充，直接从目标图像的左上角开始计算。

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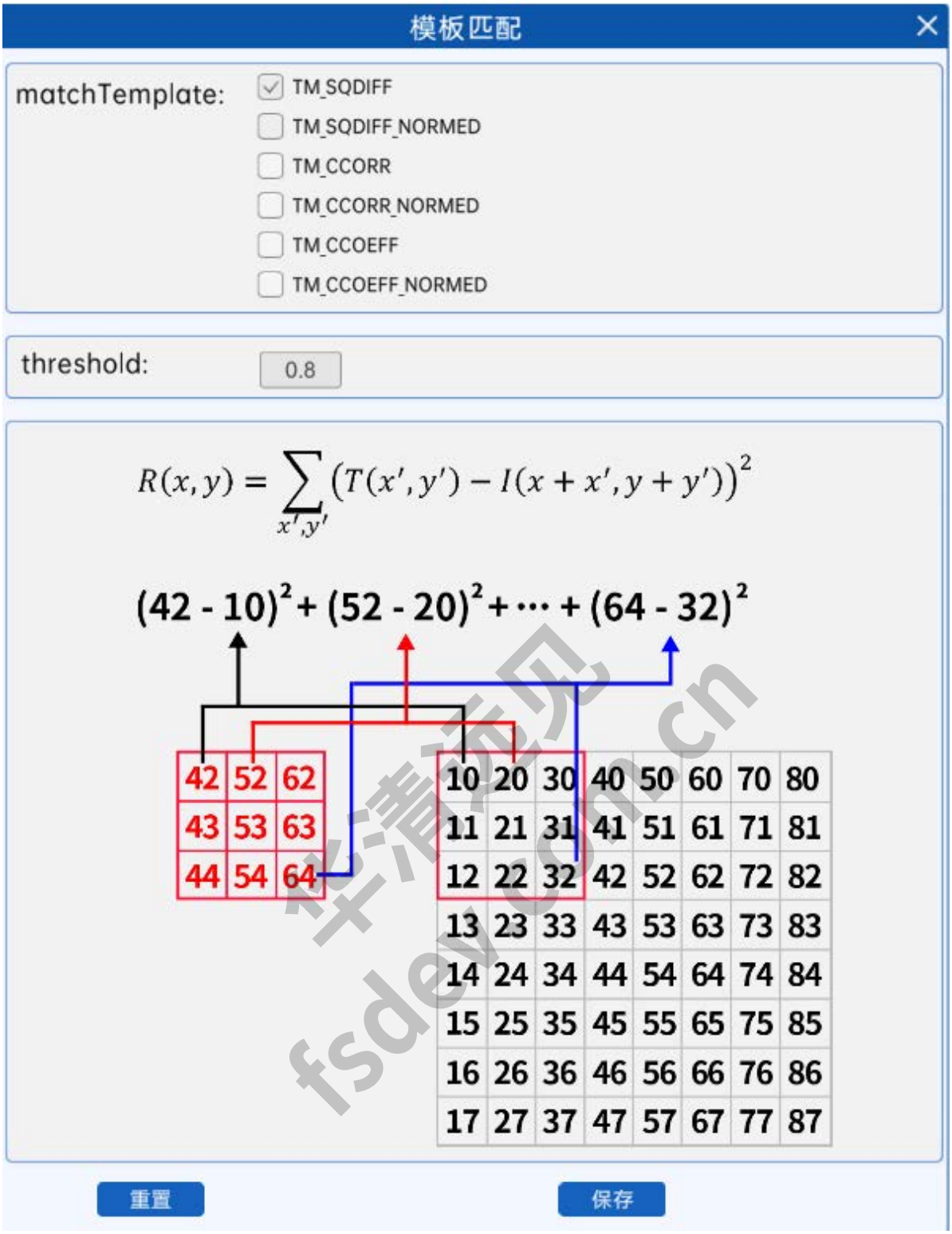

2.2 归一化平方差匹配TM_SQDIFF_NORMED

与平方差匹配类似，只不过需要将值统一到0到1，计算结果越小，代表匹配程度越高，计算过程举例如下。

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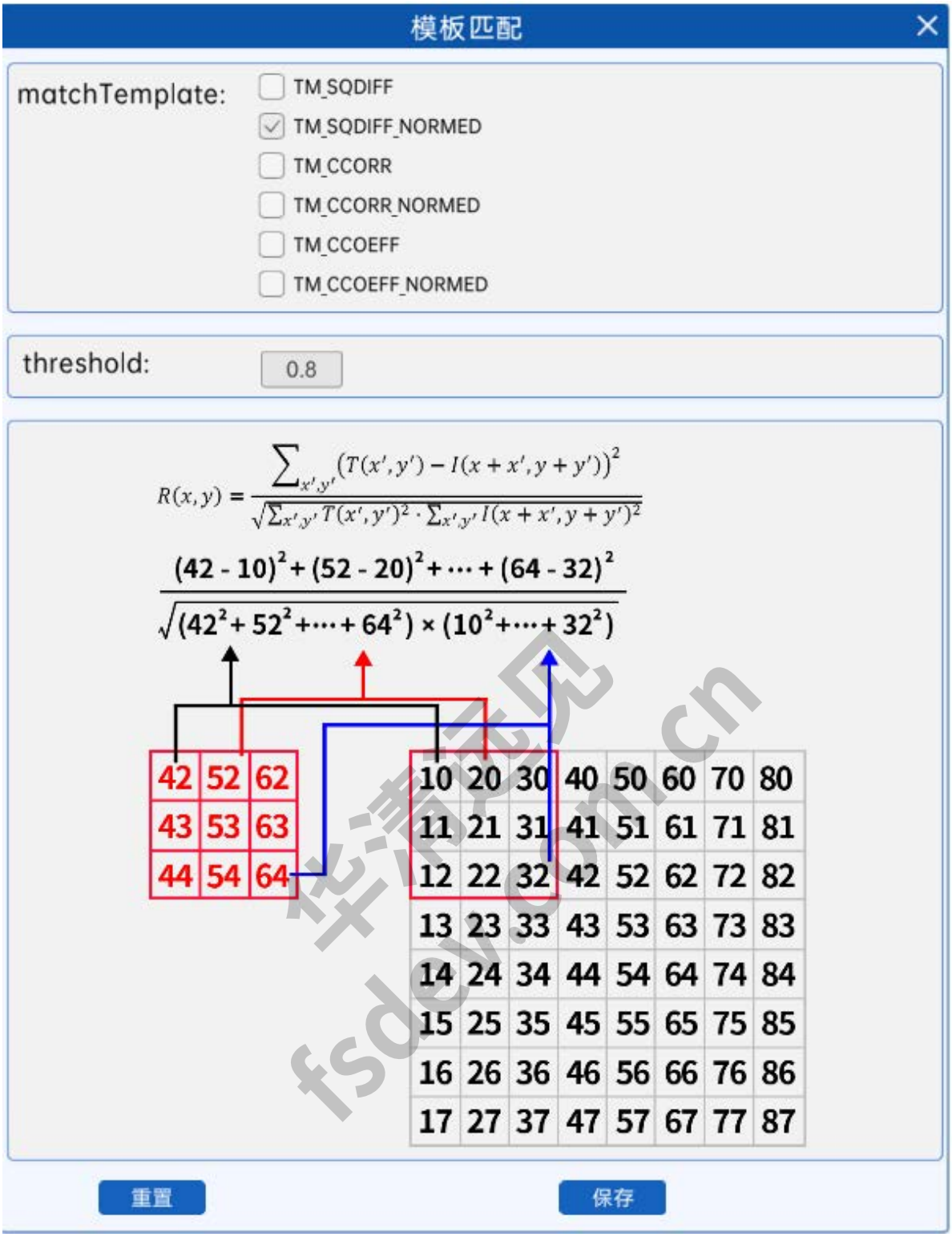

2.3 相关匹配TM_CCORR

使用对应像素的乘积进行匹配，乘积的结果越大其匹配程度越高，计算过程举例如下。

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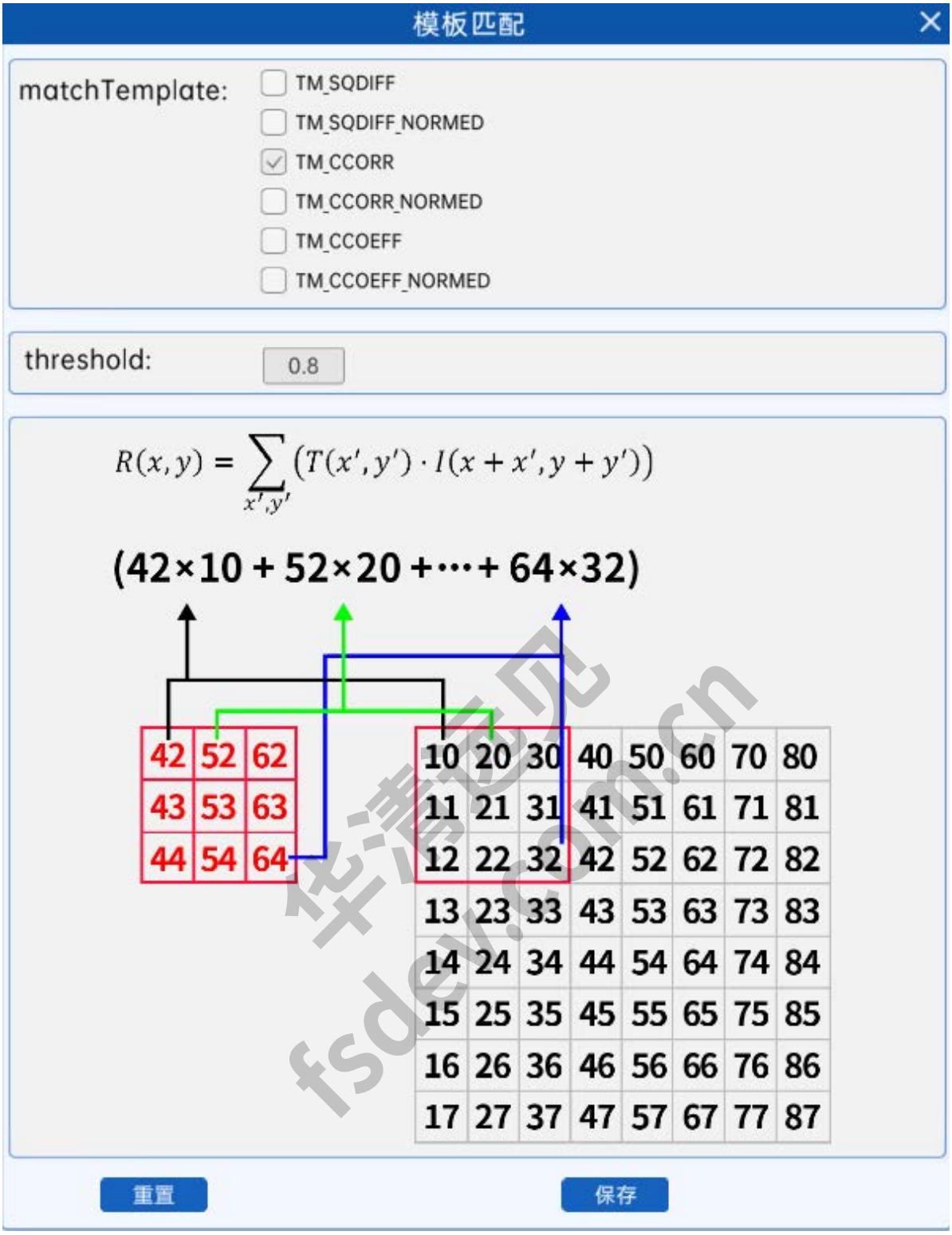

2.4 归一化相关匹配TM_CCORR_NORMED

与相关匹配类似，只不过是将其值统一到0到1之间，值越大，代表匹配程度越高，计算过程举例如下。

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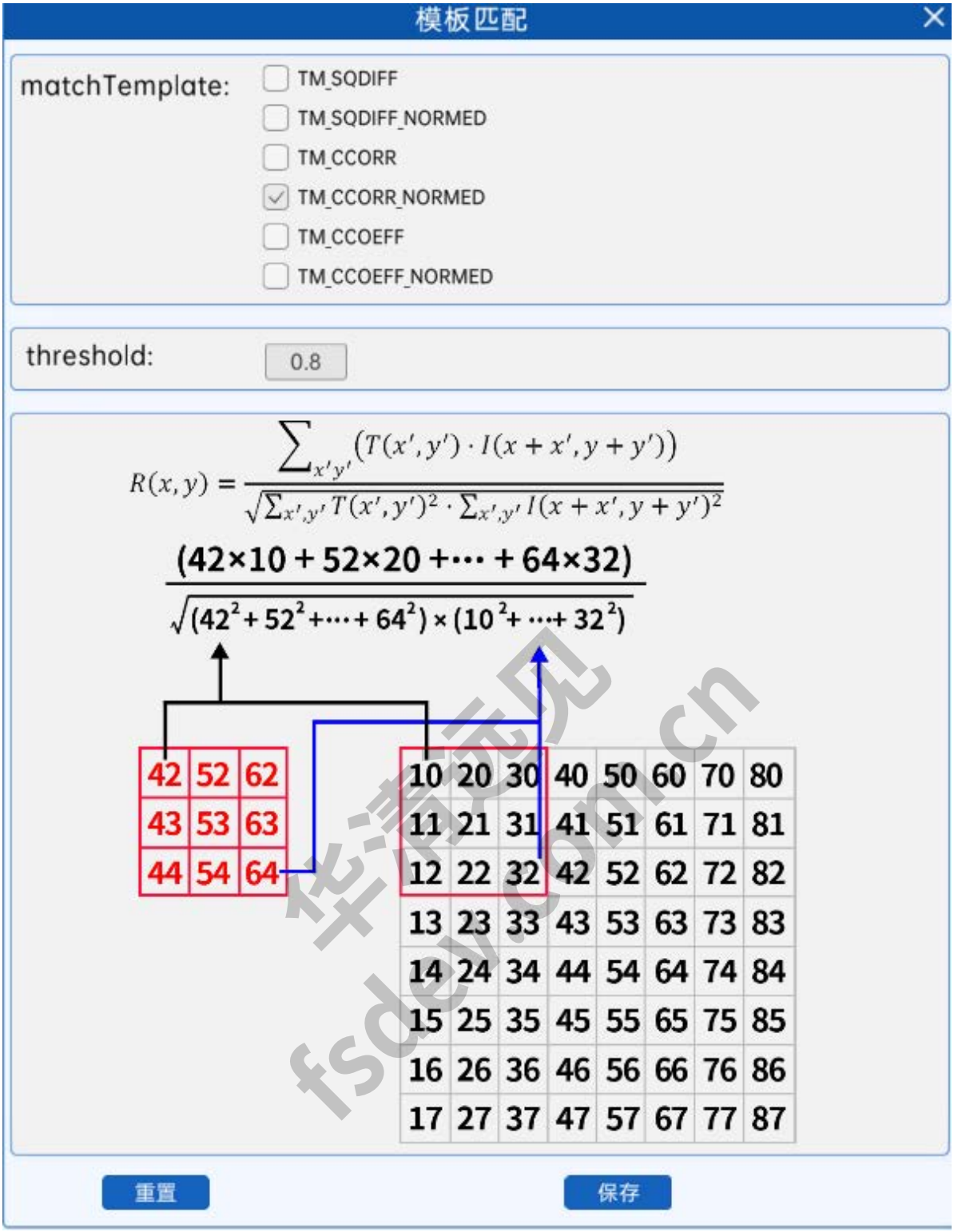

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2.5 相关系数匹配TM_CCOEFF

需要先计算模板与目标图像的均值，然后通过每个像素与均值之间的差的乘积再求和来表示其匹配程度，1表示完美的匹配，-1表示最差的匹配，计算过程举例如下。

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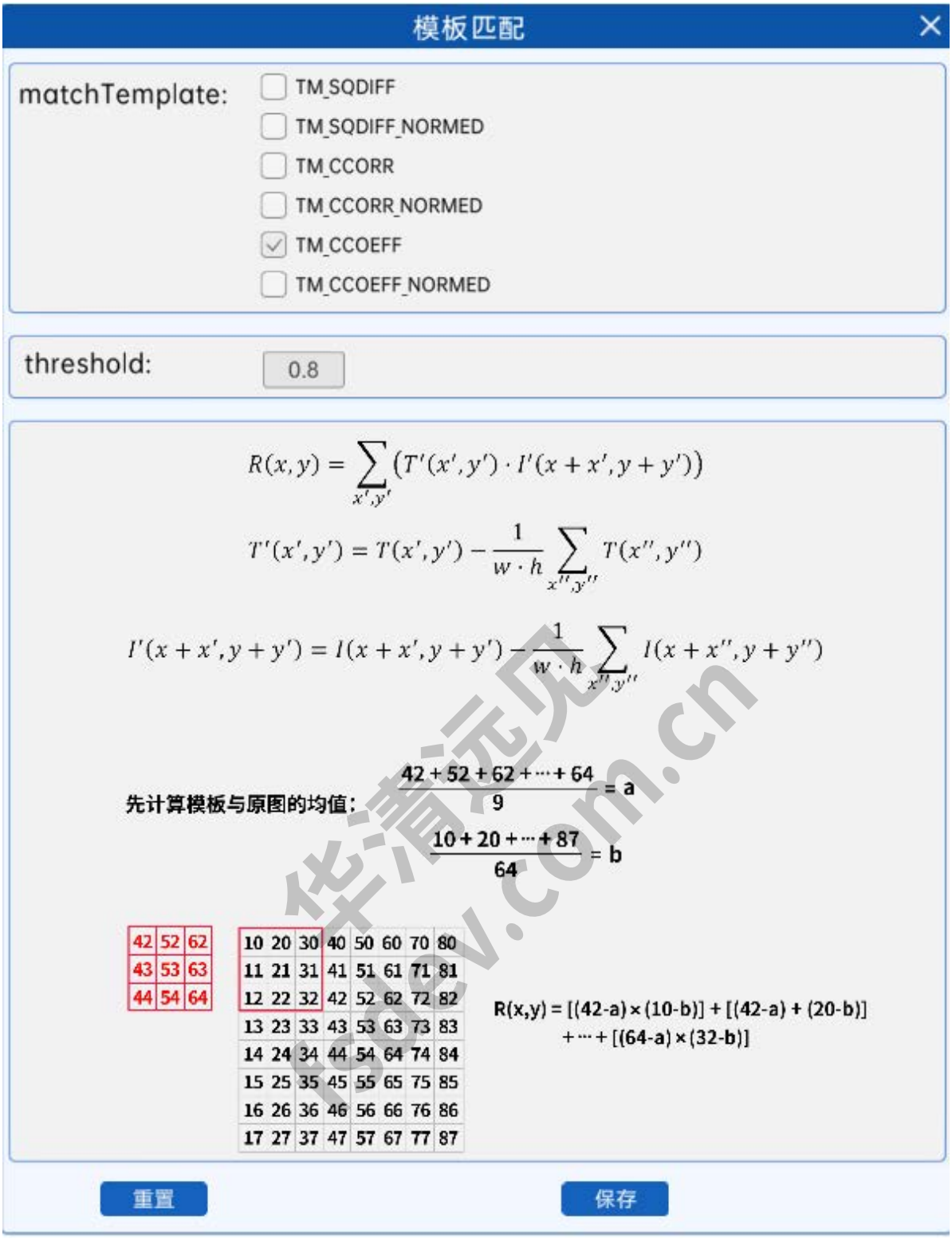

2.6 归一化相关系数匹配TM_CCOEFF_NORMED

也是将相关系数匹配的结果统一到0到1之间，值越接近1代表匹配程度越高，计算过程举例如下。

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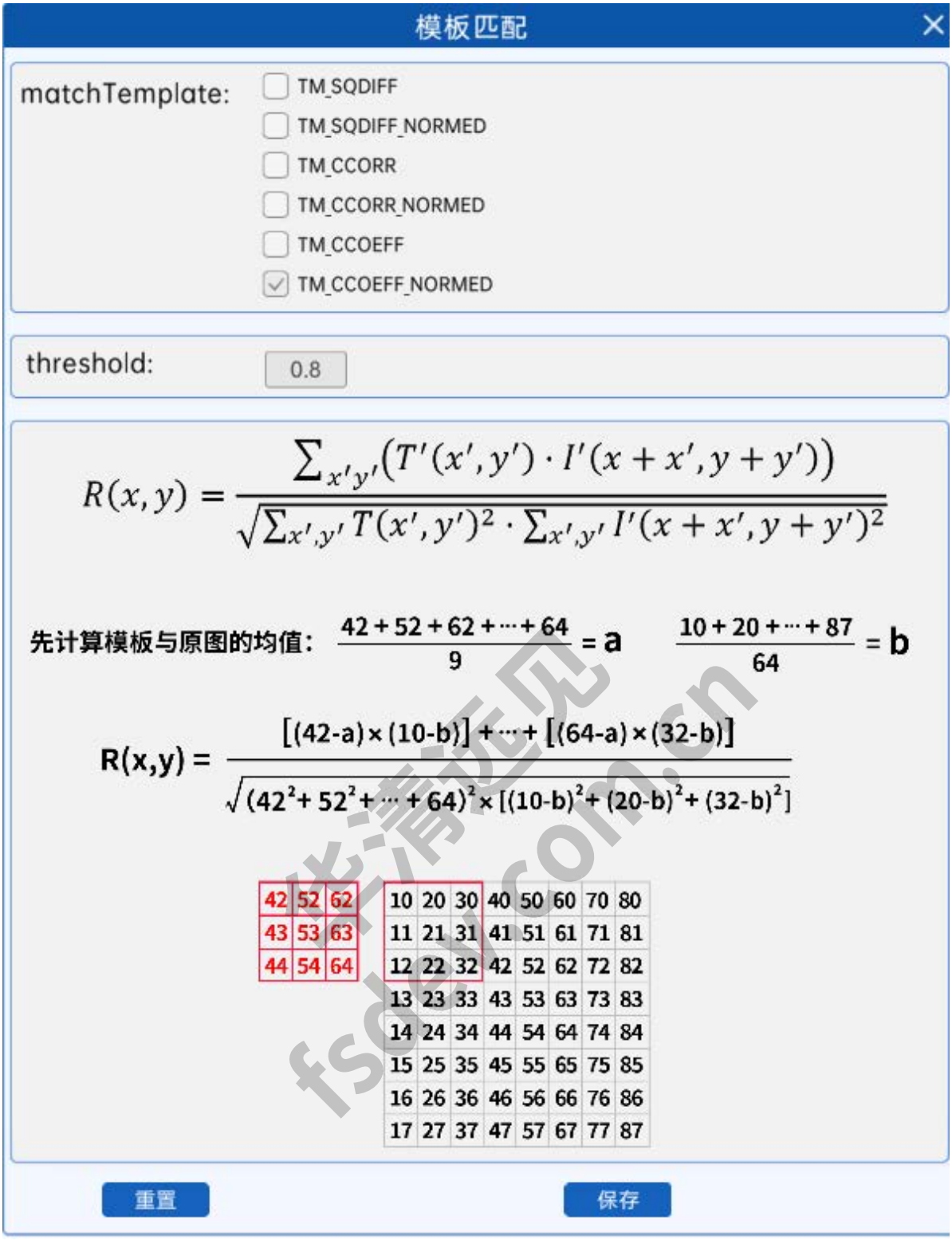

上面我们是找的目标图像中匹配程度最高的点，我们可以设定一个匹配阈值来筛选出多个匹配程度高的区域。因此在模板匹配组件中，通过matchTemplate参数来选择匹配方法，通过threshold参数来设置匹配阈值。

3. 绘制轮廓

找到符合要求的区域后，需要使用绘制轮廓组件将其在目标图像中绘制出来，通过记录下来的符合要求的区域位置与模板的宽和高，就能在目标图像中将模板框选出来。

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4. 编程实例与步骤

需要通过元宇宙平台进入“人工智能虚拟仿真实验”界面和打开“华清远见人工智能虚拟仿真本地服务管理平台”软件。

4.1 （可选）使用“华清远见人工智能虚拟仿真本地服务管理平台”启动服务

服务启动一次之后就不需要再次启动了。

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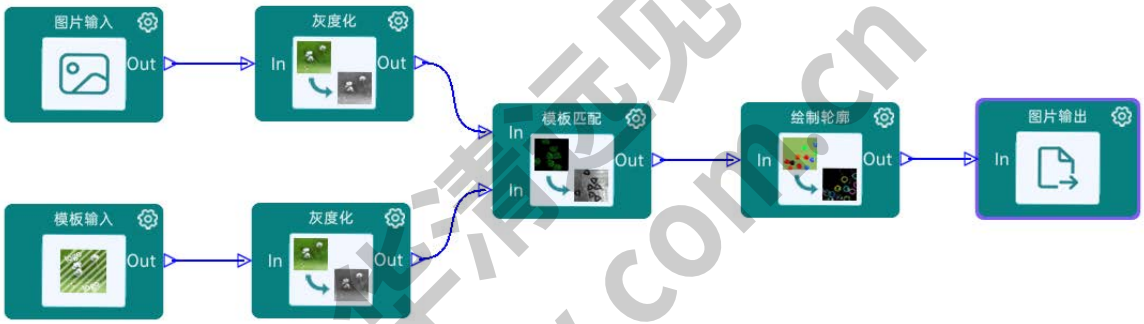
4.2 单击模板匹配实验进入

如果之前打开了进入了某个实验，可以点击“返回”回到主页即可。

4.3 拖出组件并连线

首先，我们需要模板输入组件，并对所输入的图片进行灰度化，之后将原图和模板的灰度图分别作为输入给到模板匹配组件的输入端，再将该组件的输出给到绘制轮廓的输入端，就可得到匹配后的结果图。将我们的逻辑抽象成组件之后，连线如下图所示。



4.4 验证

点击验证，如显示校验成功，即代表逻辑无误，验证按钮与校验成功界面如下图所示。

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4.5 运行

点击运行，可以实现功能运行，并显示结果输出，如下图所示。

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4.6 代码生成

点击界面左下角的生成代码按钮，可生成本实验的Python代码，生成代码按钮如下图所示。

点击后可以查看该实验代码，代码会根据界面所选参数进行适配，点击复制可以将其复制，并粘贴在其它地方。

这里我们可以粘贴到软件的Jupyter中运行，“华清远见人工智能虚拟仿真本地服务管理平台”打开Jupyter，第一次打开Jupyter需要点击“打开CMD命令”，打开Jupyter后无需再次打开CMD，如下图所示。

在弹出的CMD命令行中，输入python -m jupyterlab --notebook-dir .\\learn，点击回车，浏览器会自动跳转打开Jupyter，如下图所示。

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```
F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\workspace>python
-m jupyterlab --notebook-dir .\\learn
[I 2022-12-26 09:28:37.282 ServerApp] jupyterlab | extension was successfully linked.
[W 2022-12-26 09:28:37.285 NotebookApp] 'shutdown_no_activity_timeout' has moved from NotebookApp to ServerApp. This con
fig will be passed to ServerApp. Be sure to update your config before our next release.
[I 2022-12-26 09:28:37.288 ServerApp] nbclassic | extension was successfully linked.
[I 2022-12-26 09:28:37.594 ServerApp] notebook_shim | extension was successfully linked.
[I 2022-12-26 09:28:37.613 ServerApp] notebook_shim | extension was successfully loaded.
[I 2022-12-26 09:28:37.613 LabApp] JupyterLab extension loaded from F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码\BS
架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\tools\Python\lib\site-packages\jupyterlab
[I 2022-12-26 09:28:37.613 LabApp] JupyterLab application directory is F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码
\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\tools\Python\share\jupyter\lab
[I 2022-12-26 09:28:37.616 ServerApp] jupyterlab | extension was successfully loaded.
[I 2022-12-26 09:28:37.623 ServerApp] nbclassic | extension was successfully loaded.
[I 2022-12-26 09:28:37.624 ServerApp] Serving notebooks from local directory: F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿
真\源码\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\workspace\learn
[I 2022-12-26 09:28:37.624 ServerApp] Jupyter Server 1.23.4 is running at:
[I 2022-12-26 09:28:37.624 ServerApp] http://localhost:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9ee7
[I 2022-12-26 09:28:37.624 ServerApp]  or http://127.0.0.1:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9e
e7
[I 2022-12-26 09:28:37.624 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirm
ation).
[C 2022-12-26 09:28:37.646 ServerApp]

    To access the server, open this file in a browser:
        file:///C:/Users/82553/AppData/Roaming/jupyter/runtime/jpserver-27004-open.html
    Or copy and paste one of these URLs:
        http://localhost:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9ee7
```

如果没有自动跳转和打开浏览器，请输入下图中圈出的内容，每个人的该内容不一致，请复制自己的内容。

```
F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\workspace>python
-m jupyterlab --notebook-dir .\\learn
[I 2022-12-26 09:28:37.282 ServerApp] jupyterlab | extension was successfully linked.
[W 2022-12-26 09:28:37.285 NotebookApp] 'shutdown_no_activity_timeout' has moved from NotebookApp to ServerApp. This con
fig will be passed to ServerApp. Be sure to update your config before our next release.
[I 2022-12-26 09:28:37.288 ServerApp] nbclassic | extension was successfully linked.
[I 2022-12-26 09:28:37.594 ServerApp] notebook_shim | extension was successfully linked.
[I 2022-12-26 09:28:37.613 ServerApp] notebook_shim | extension was successfully loaded.
[I 2022-12-26 09:28:37.613 LabApp] JupyterLab extension loaded from F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码\BS
架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\tools\Python\lib\site-packages\jupyterlab
[I 2022-12-26 09:28:37.613 LabApp] JupyterLab application directory is F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码
\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\tools\Python\share\jupyter\lab
[I 2022-12-26 09:28:37.616 ServerApp] jupyterlab | extension was successfully loaded.
[I 2022-12-26 09:28:37.623 ServerApp] nbclassic | extension was successfully loaded.
[I 2022-12-26 09:28:37.624 ServerApp] Serving notebooks from local directory: F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿
真\源码\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\workspace\learn
[I 2022-12-26 09:28:37.624 ServerApp] Jupyter Server 1.23.4 is running at:
[I 2022-12-26 09:28:37.624 ServerApp] http://localhost:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9ee7
[I 2022-12-26 09:28:37.624 ServerApp]  or http://127.0.0.1:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9e
e7
[I 2022-12-26 09:28:37.624 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirm
ation).
[C 2022-12-26 09:28:37.646 ServerApp]

    To access the server, open this file in a browser:
        file:///C:/Users/82553/AppData/Roaming/jupyter/runtime/jpserver-27004-open.html
    Or copy and paste one of these URLs:
        http://localhost:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9ee7
```

将代码输入到Jupyter中，注意这里需要往Jupyter导入一张demo.png，在大多数情况下，这张照片最好不要是其它后缀修改成png后缀而来，而是直接后缀就是png。点击运行可以查看结果。

5. 实验现象

可以看到，不同的参数对于模板匹配的结果所产生的不同，实验结果为：

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```python
import cv2
import numpy as np

if __name__ == "__main__":
    path_search = "./images/maliao.png"
    image_np = cv2.imread(path_search)
    image_np_gray = cv2.cvtColor(image_np, cv2.COLOR_BGR2GR
    path_target = "./images/shuiyin.png"
    template = cv2.imread(path_target)
    template_gray = cv2.cvtColor(template, cv2.COLOR_BGR2GR
    h, w = template_gray.shape[:2]
    res = cv2.matchTemplate(image_np_gray, template_gray, c
    threshold = 0.8
    loc = np.where(res >= threshold)  # 匹配程度大于threshold的坐标y,x
    for pt in zip(*loc[::-1]):  # *号表示可选参数
        right_bottom = (pt[0] + w, pt[1] + h)
        cv2.rectangle(image_np, pt, right_bottom, (0, 0, 255), 2)
    # 返回处理正确后的内容
    cv2.imshow("image_np", image_np)
    cv2.waitKey(0)
```

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